Answer key. Answers are in magenta.

NRES 746 – Week 4, Lecture 2: Expectation and the Sample Mean

Instructions: Work alone for about 8-10 minutes. Then compare with a neighbor.

Part 1: A parameter vs. a statistic

Suppose the length of brook trout in a stream (X) follows some distribution ($P(X = x) = f(x)$) with true mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \mathbb{E}[(X - \mu)^2]$ – these describe the *entire population* of trout in the stream, whether or not anyone ever catches one.

You catch $n = 5$ trout, measure them, and compute their average length $\bar{x}$.

1. Is μ a random variable? Is $\bar{x}$ a random variable? Briefly explain your reasoning.

μ is not a random variable: it's a fixed (unknown) property of the whole population. $\bar{x}$ is: a different sample of 5 trout would give a different value.

Part 2: working with expectations

Here are some useful facts, true for any random variable X and constants a, b:

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b \qquad \text{Var}(aX) = a^2\text{Var}(X)$$

Also, for any two random variables X and Y:

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

One more thing: the variance of any constant is zero.

Suppose trout length X has $\mathbb{E}[X] = \mu = 22$ cm and $\text{Var}(X) = \sigma^2 = 16$ cm^2.

2a. What is $\mathbb{E}[2X + 3]$? What is $\text{Var}(2X + 3)$?

$\mathbb{E}[2X] = 2(22) = 44$, so $\mathbb{E}[2X + 3] = 47$ cm $\qquad$ $\text{Var}(2X) = 2^2(16) = 64$ cm^2, and $\text{Var}(2X + 3) = 64$ cm^2 too

2b. A second, independent measurement Y of a different trout has the same mean and variance as X. It is always true that $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$. It is also true in this case that $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$. However, this second fact is not always true. What do you think might be the special condition that makes it true?

X and Y are independent (strictly, uncorrelated is enough).

Part 3: Averaging two measurements

Let X_1 and X_2 be two independent measurements of the same quantity, each with mean μ and variance σ^2. Define their average $M = \dfrac{X_1 + X_2}{2}$.

3a. Rewrite M as $aX_1 + bX_2$ (i.e. what are a and b?), then use the facts from Part 2 to derive $\mathbb{E}[M]$ and $\text{Var}(M)$ in terms of μ and σ^2. Show every step.

$a = b = \frac{1}{2}$. $\quad \mathbb{E}[M] = \frac{1}{2}\mu + \frac{1}{2}\mu = \mu, \qquad \text{Var}(M) = \left(\frac{1}{2}\right)^2\sigma^2 + \left(\frac{1}{2}\right)^2\sigma^2 = \dfrac{\sigma^2}{2}$

3b. Using $\mu = 22$ and $\sigma^2 = 16$ (from Part 2), what is $\mathrm{Var}(M)$ as a number? Compare it to $\mathrm{Var}(X_1) = 16$ alone – how does the variance of an average of two measurements compare to the variance of each of its constituent parts?

$\mathrm{Var}(M) = 16/2 = 8$ cm^2, half the variance of a single measurement

Part 4: Generalizing to n measurements

NOTE: We will do part 4 together on the whiteboard in a minute, but see if you can think this one through if you have time:

Let $X_1, X_2, \ldots, X_n$ be independent measurements, each with mean μ and variance σ^2, and define the sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$.

4a. Following the same logic as Part 3 (now with n terms instead of 2), derive $\mathbb{E}[\bar{X}]$ and $\mathrm{Var}(\bar{X})$ in terms of μ, σ^2, and n.

$$\mathbb{E}[\bar{X}] = \sum_{i=1}^{n} \frac{1}{n}\mu = \mu, \qquad \mathrm{Var}(\bar{X}) = \sum_{i=1}^{n} \frac{1}{n^2}\sigma^2 = \frac{\sigma^2}{n}$$

4b. Check your formula against $n = 5$, $\sigma^2 = 16$: what number does $\mathrm{Var}(\bar{X})$ come out to? How does it compare to Part 3b when you plug in $n = 2$?

$\mathrm{Var}(\bar{X}) = 16/5 = 3.2$ cm^2; with $n = 2$ it gives 8, matching 3b

4c. As $n \to \infty$, what happens to $\mathrm{Var}(\bar{X})$? What does that mean, in plain language, about how precisely you can pin down μ with a very large sample?

$\sigma^2/n \to 0$: with a large enough sample, $\bar{x}$ estimates μ with as little uncertainty as you like.

4d. One more thing to be careful about: $\mathrm{Var}(\bar{X})$ and the *standard deviation* of $\bar{X}$ (often called the standard error) are not the same formula. If $\mathrm{Var}(\bar{X}) = \sigma^2/n$, write down the standard deviation of $\bar{X}$ – how does it scale with n?

$\mathrm{SD}(\bar{X}) = \sigma/\sqrt{n}$; it shrinks like $1/\sqrt{n}$ (here $4/\sqrt{5} \approx 1.79$ cm).